

# Measures of Central Tendency

- ❖ Measures of central tendency are statistical measures that describe the center or average of a set of data values. They provide a single value that represents the central or typical value in a dataset. The three main measures of central tendency are:

## Mean:

- ❖ The mean, often referred to as the average, is calculated by **adding up all the values** in a dataset and then **dividing by the number of values**.
- ❖ Formula:

$$m = \frac{\text{sum of the terms}}{\text{number of terms}}$$

$m$  = mean

## **Median:**

- ❖ The median is the middle value in a dataset when it is ordered from least to greatest. If there is an even number of values, the median is the average of the two middle values.
- ❖ To find the median, the data must be sorted first.
- ❖ For an odd number of values: Median = Middle value
- ❖ For an even number of values: Median = Sum of two middle values / 2

## **Mode:**

- ❖ The mode is the value that appears most frequently in a dataset.
- ❖ A dataset may have no mode (if all values occur with the same frequency), one mode (unimodal), or multiple modes (multimodal).

# Dispersion

- ❖ Dispersion measures in statistics quantify the extent to which data values spread out or deviate from the central tendency. They provide insights into the variability, or spread, within a dataset. Common measures of dispersion include:

## Range:

- ❖ The range is the simplest measure of dispersion and is calculated as the difference between the maximum and minimum values in a dataset.
- ❖ **Formula: Range = Maximum value – Minimum value**

## Variance:

- ❖ Variance measures the average squared deviation of each data point from the mean of the dataset.

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N} \quad \text{Population Variance}$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} \quad \text{Sample Variance}$$